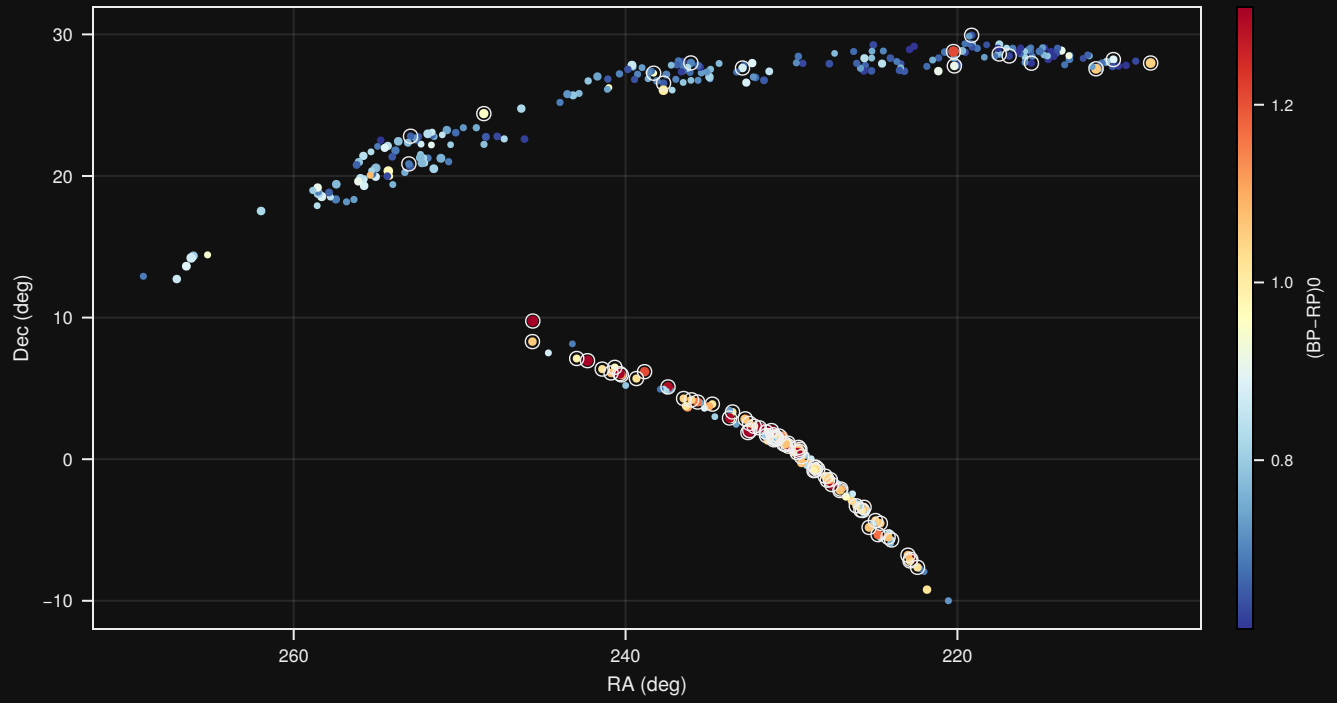
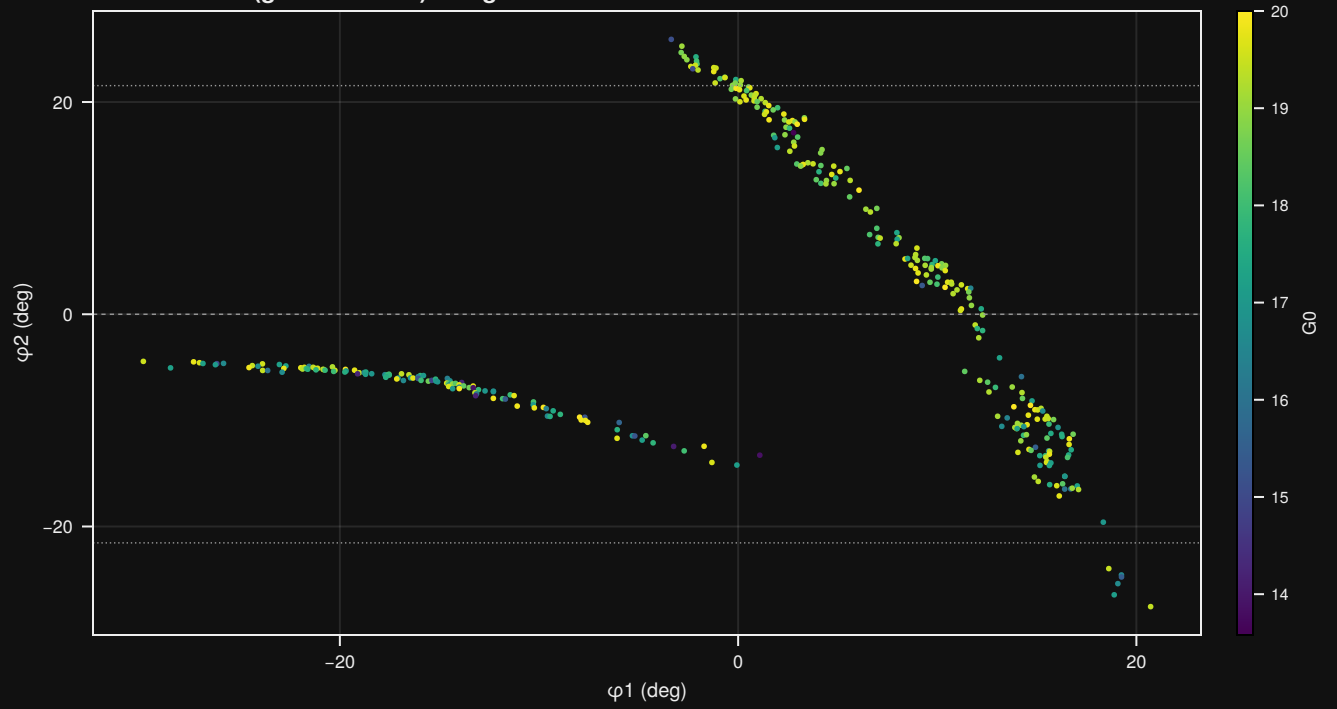
Pal 5

MEMBERSHIP	
Stars in catalog	363
with Gaia GSP-Phot	181 (50%)
with radial velocity	85
Stream id (sID)	61, 64
GEOMETRY · great-circle fit	
Centroid RA, Dec	233.9, 21.7 deg
Centroid l, b	28.4, 45.8 deg
Pole RA, Dec	132.0, 34.4 deg
Length (φ1, 2–98%)	42.6 deg
Width σ(φ2), robust	10.77 deg
KINEMATICS	
pmRA* med ± σ	0.87 ± 4.98 mas/yr
pmDE med ± σ	-3.64 ± 1.61 mas/yr
μφ1* med ± σ	-2.86 ± 0.78 mas/yr
μφ2 med ± σ	-2.50 ± 4.66 mas/yr
Parallax (cat.) med	0.101 ± 0.243 mas
Parallax (wtd, ZP-corr.)	0.109 ± 0.011 mas
Distance	d = 9.2 ± 1.0 kpc
Physical length	6.83 kpc
RV median ± σ(MAD)	-57.8 ± 7.0 km/s
RV gradient dV/dφ1	-3.42 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.6–20.0 (18.8)
(BP−RP)0 median	0.79
GSP-Phot [M/H] (n=181)	-1.47 ± 0.63
GSP-Phot Teff median	5414 K
RUWE > 1.4	0.0 %
PM clusters (largest)	12 (131 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	363 | 129
n_RV catalog | paper	85 | 69
<[Fe/H]> paper	-1.41
Stellar mass (paper)	17000 Msun
Pole offset vs paper	20.5 deg
LABEL CHECK	n differs from paper!

Sky map · catalog members (n = 363) · size ∝ brightness · ring = has RV

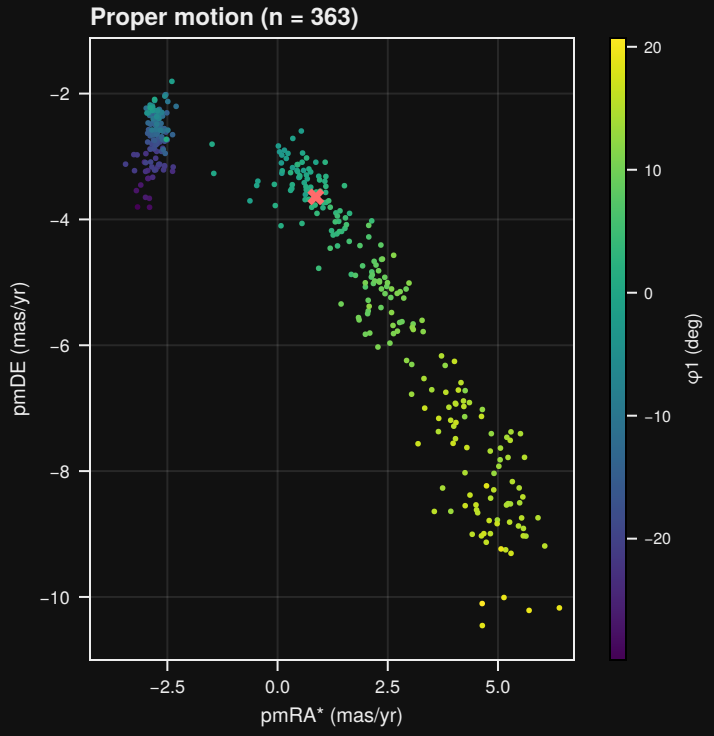


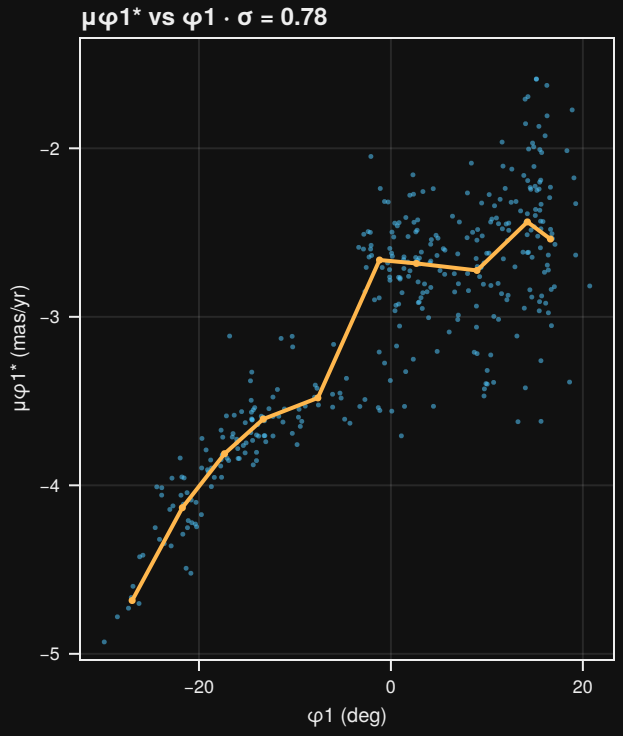
Stream frame (great-circle fit) · length 42.6° · width σ 10.77°

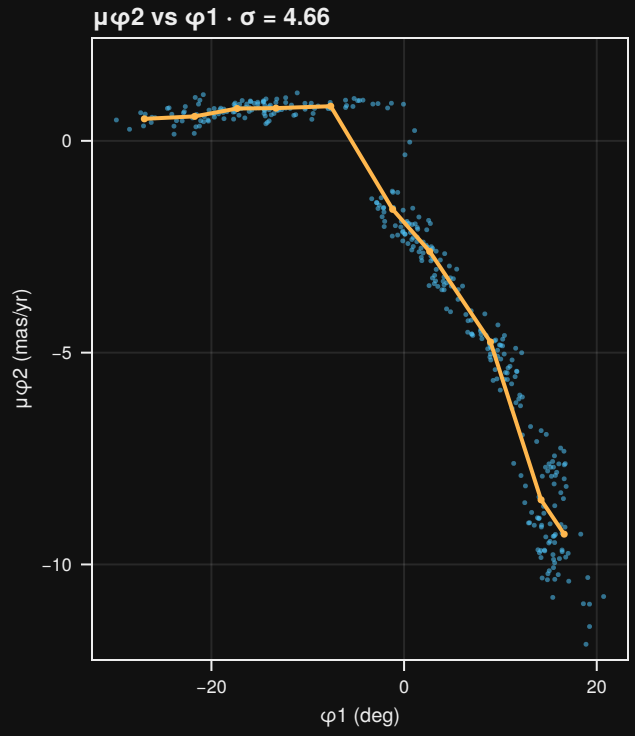


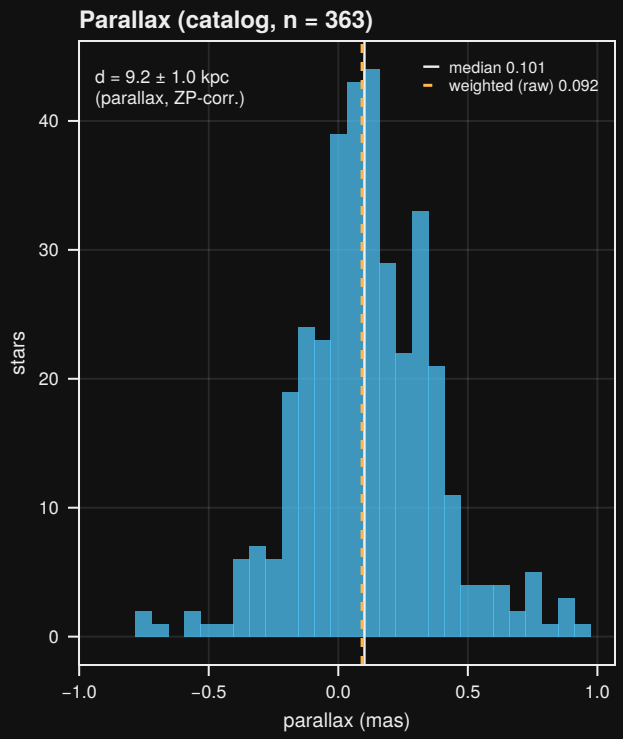
Pal 5

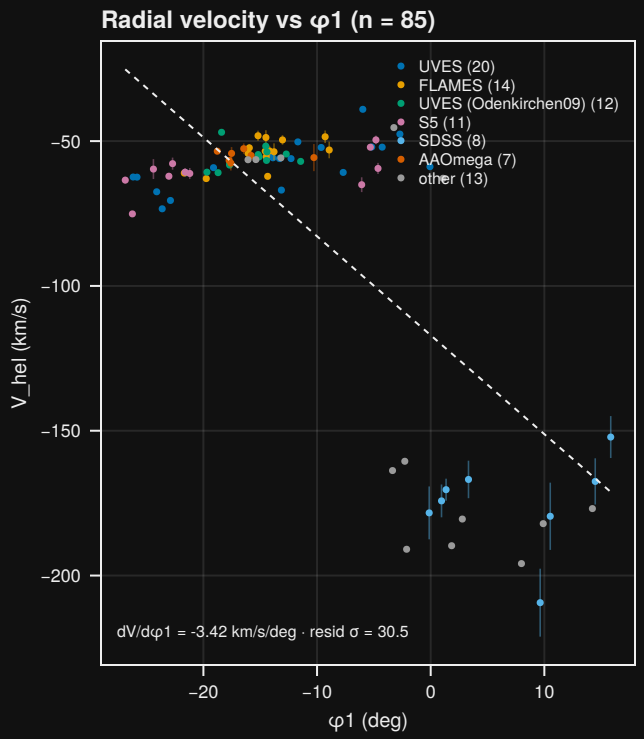
Kinematics · proper motion · parallax · radial velocity

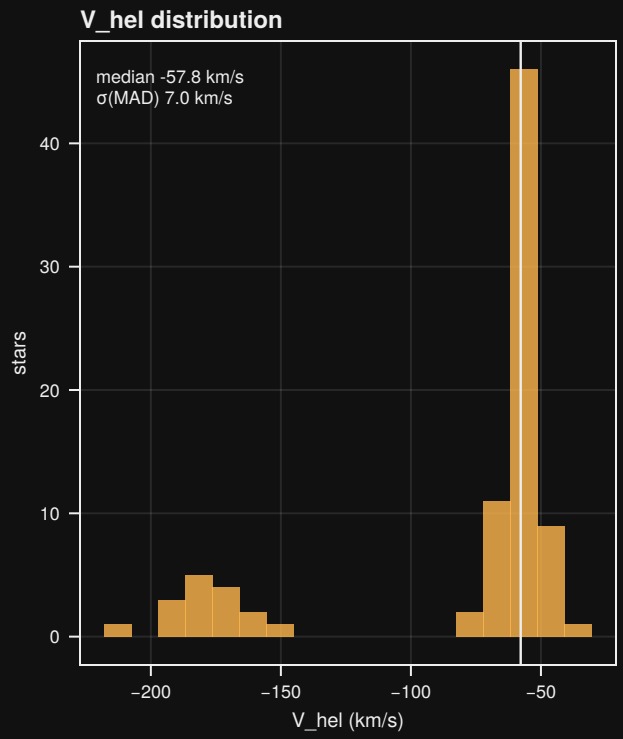






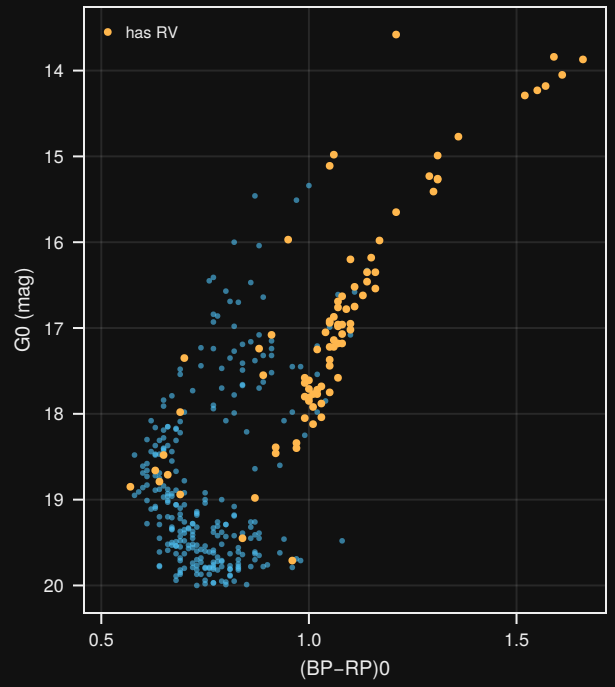




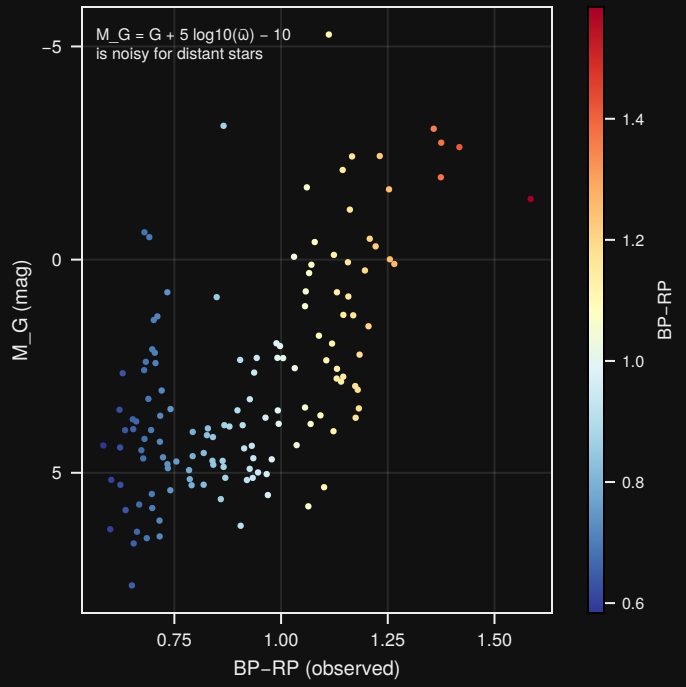


Pal 5

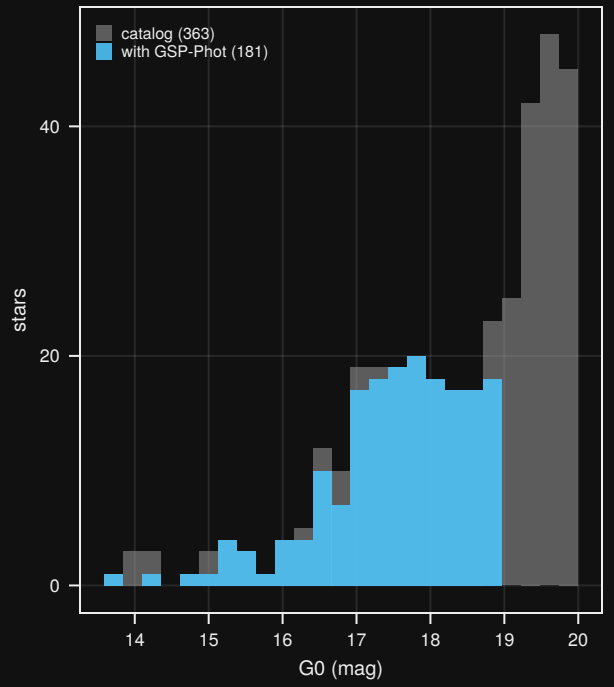
CMD · dereddened (n = 363)



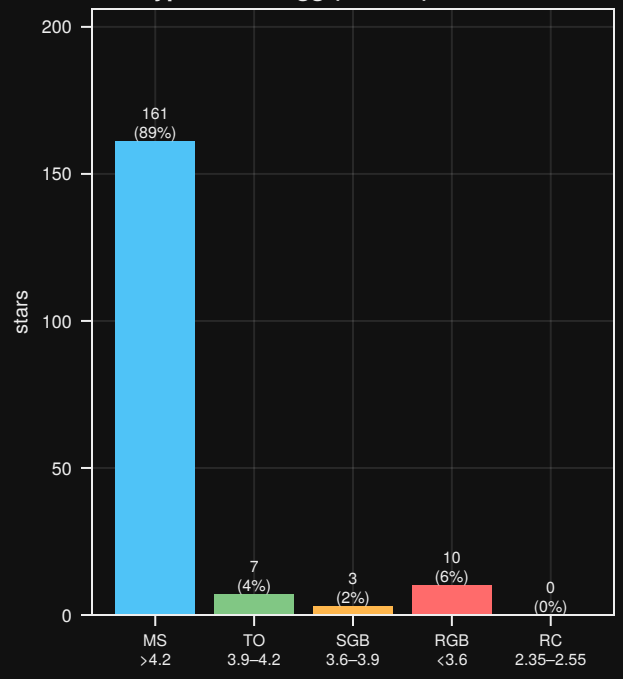
HR diagram · ϖ>0 (n = 139/181)



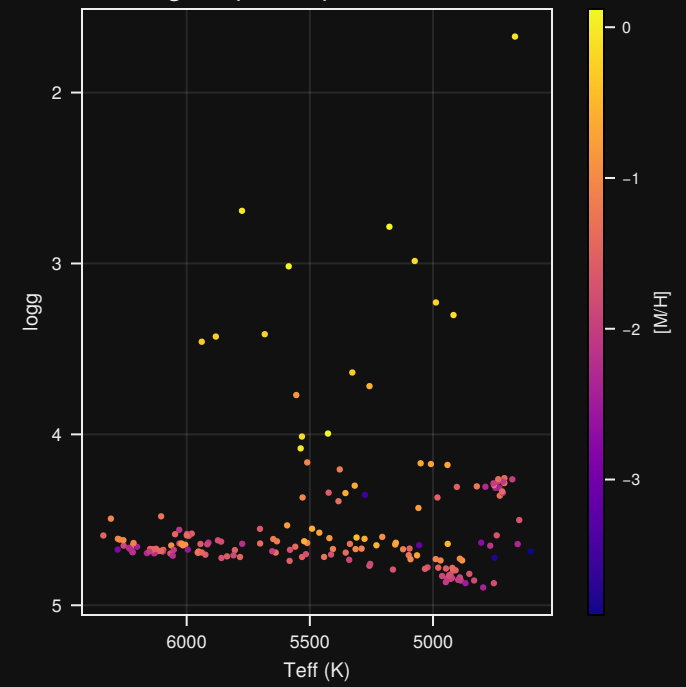
G0 & GSP-Phot coverage



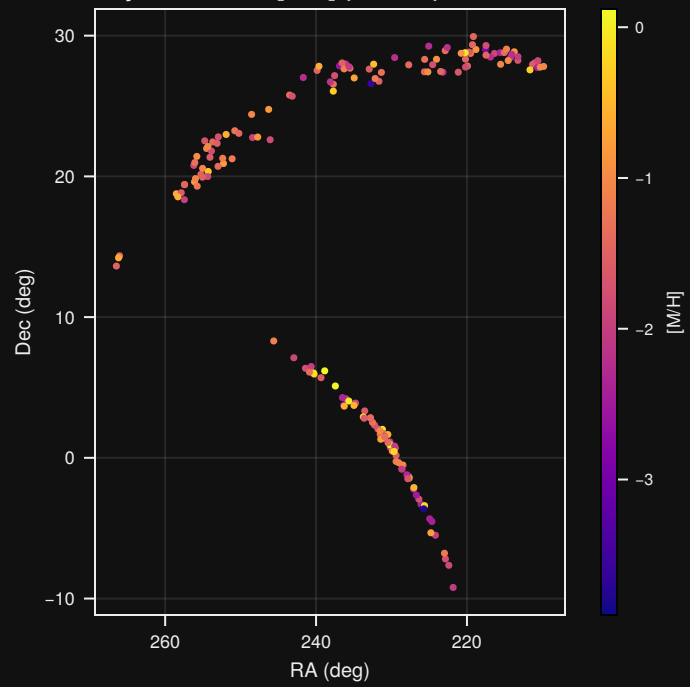
Star types from logg (n = 181)



Kiel diagram (n = 181)



Sky · GSP-Phot [M/H] (n = 181)

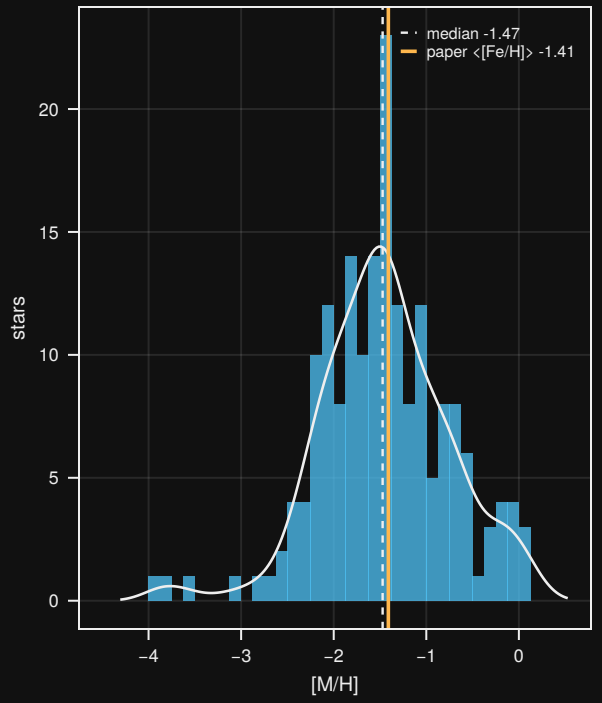


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

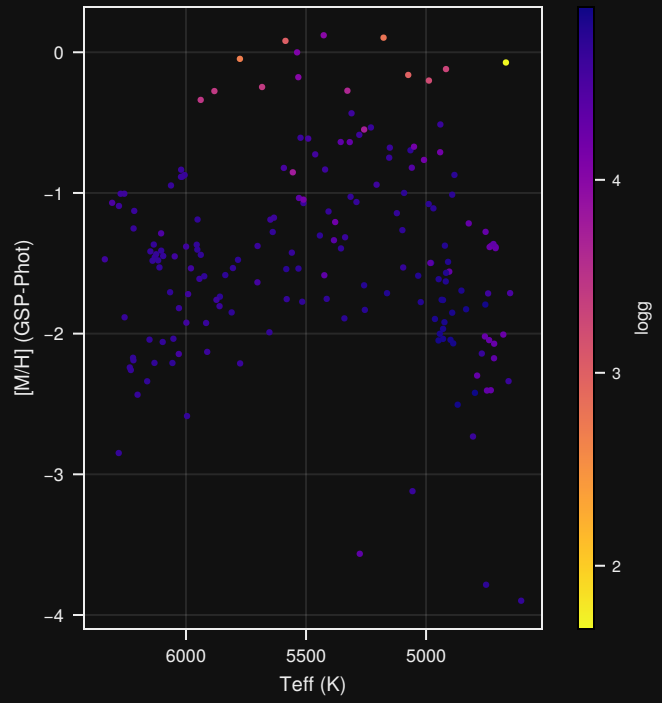
Pal 5

Metallicity · age · astrometric quality

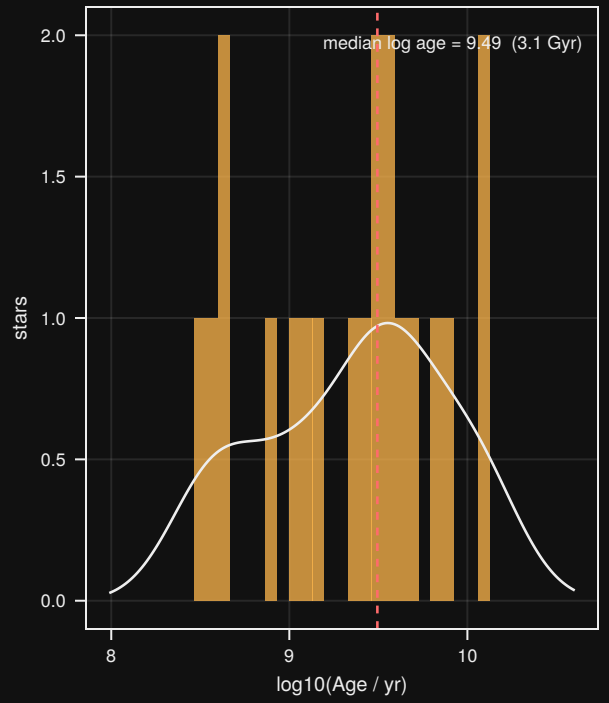
[M/H] GSP-Phot (n = 181)



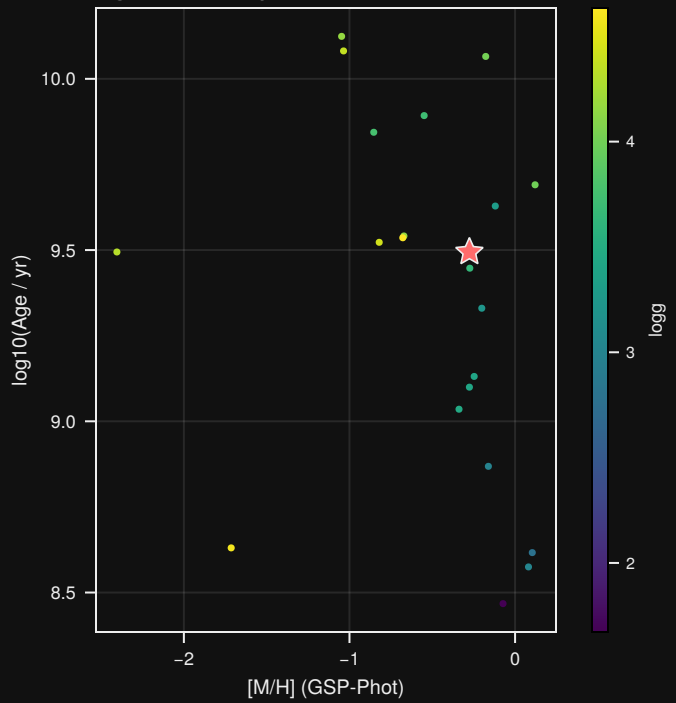
Teff vs [M/H] (n = 181)



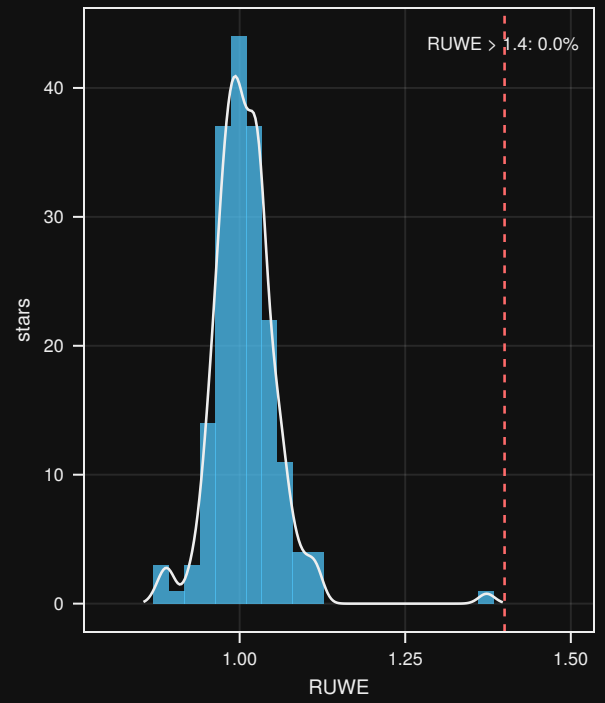
FLAME age (n = 21)



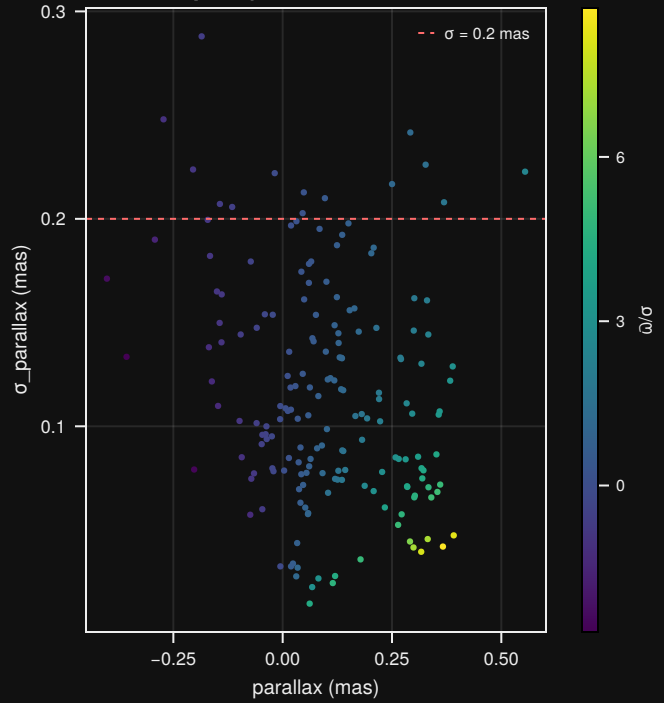
Age–metallicity (n = 21)



RUWE (n = 181)

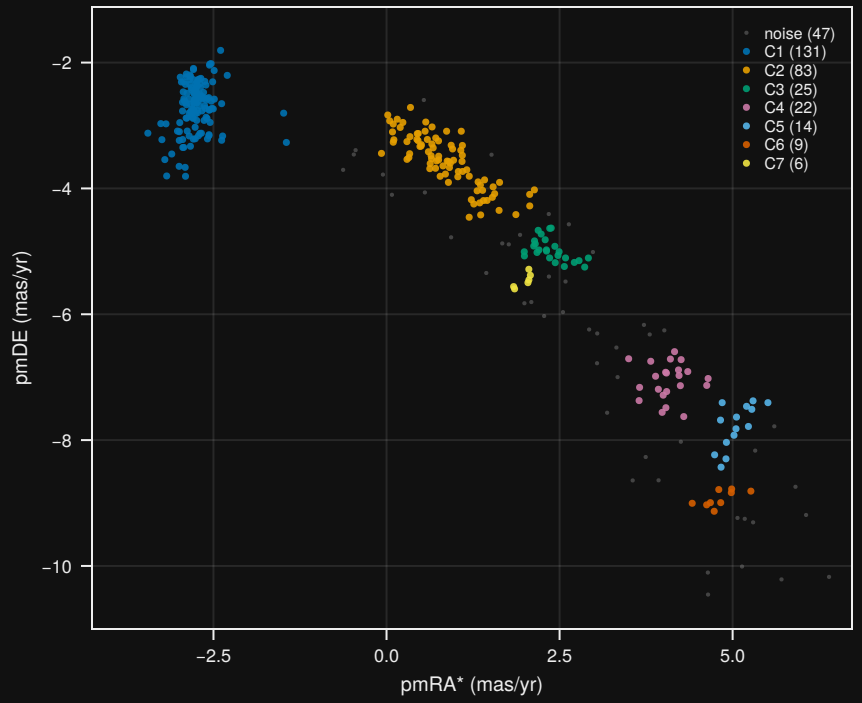


Parallax quality (n = 181)

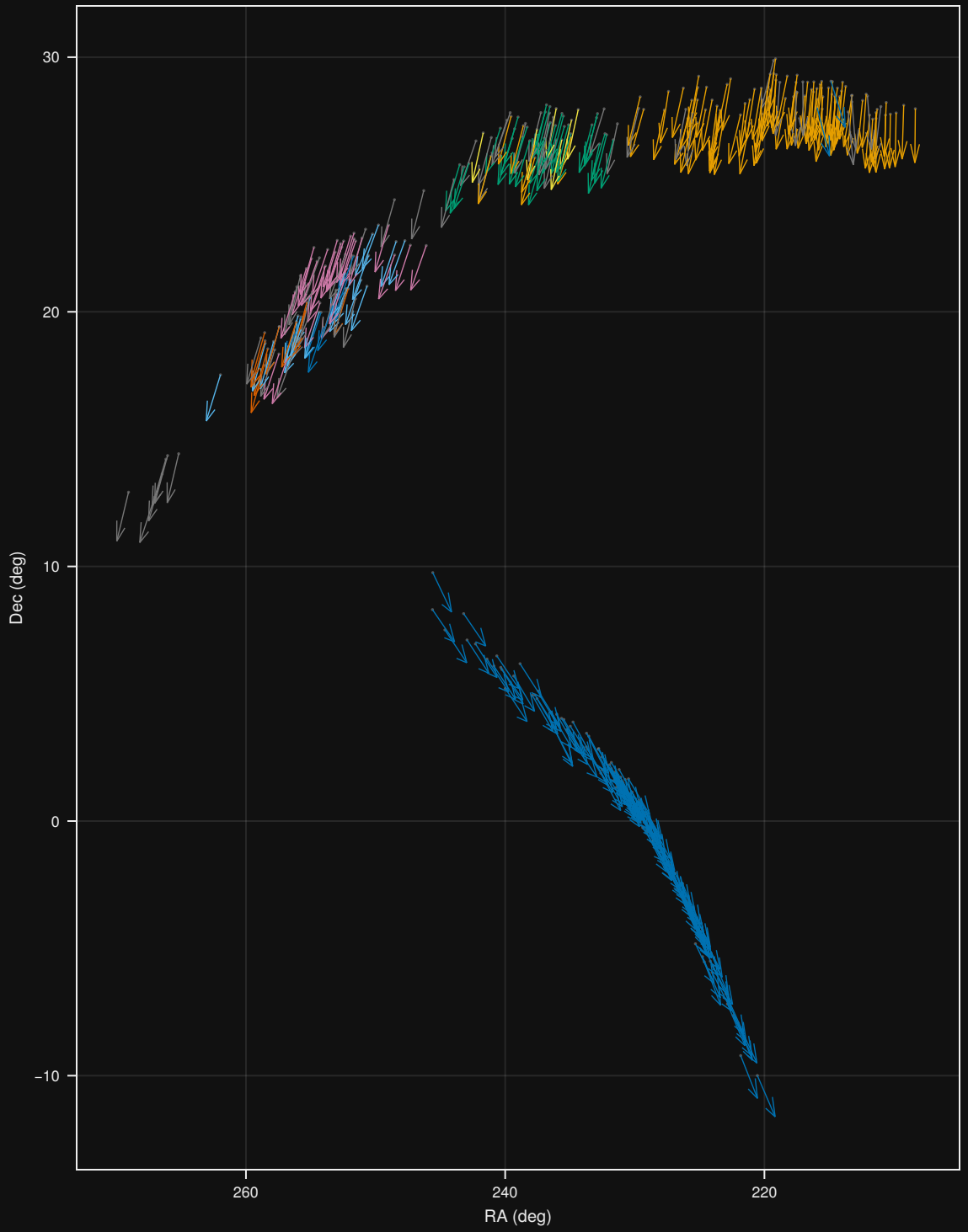


Pal 5

PM clustering · HDBSCAN · 12 cluster(s) · noise 13%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	47	3.04	-6.24	1.89	-1.13	2
C1	131	-2.76	-2.64	0.17	-1.58	69
C2	83	0.82	-3.54	0.53	-1.48	9
C3	25	2.36	-5.01	0.24	-1.54	2
C4	22	4.05	-7.00	0.26	-1.62	1
C5	14	5.03	-7.73	0.28	-1.26	0
C6	9	4.80	-8.99	0.26	-0.82	0
C7	6	2.05	-5.48	0.03	-2.14	1
C8	6	5.38	-8.51	0.20	—	0

Pal 5

Per-cluster proper-motion vector maps

